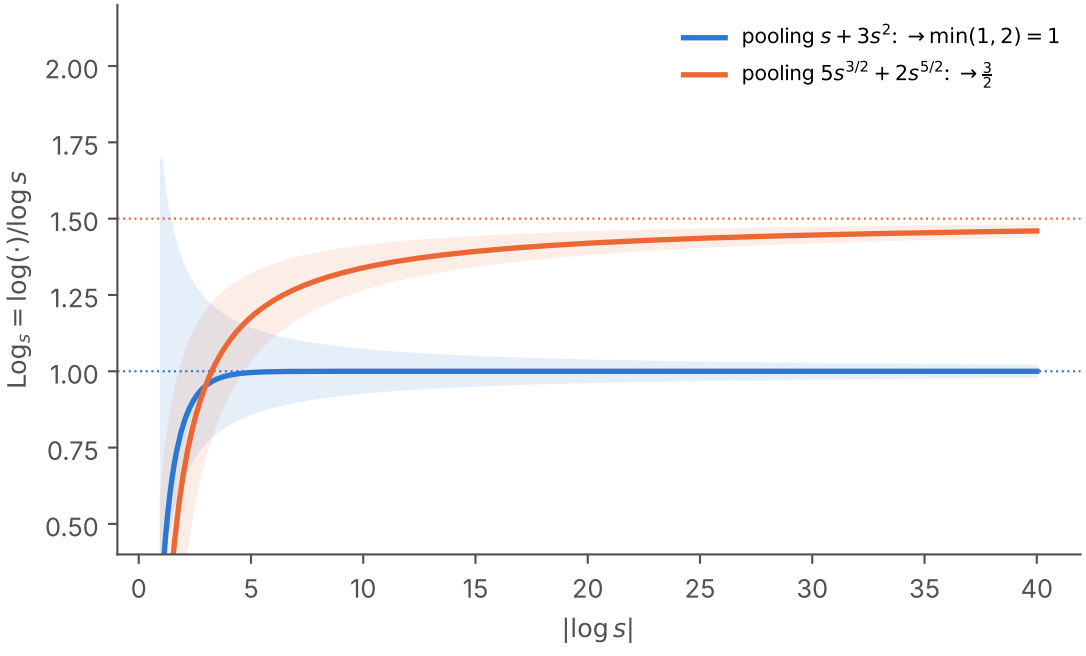
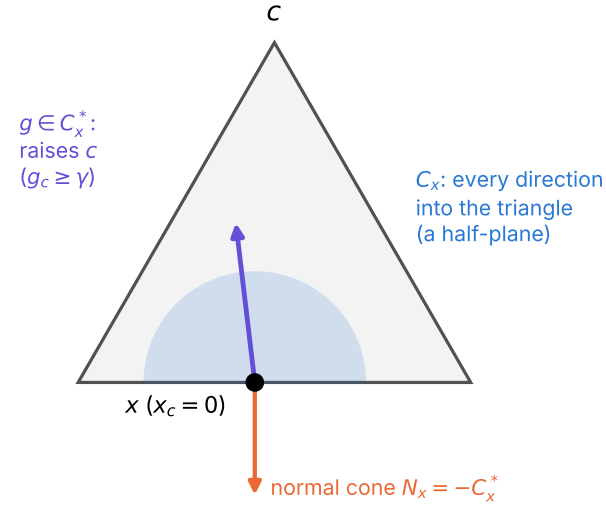
(a) under the log map pooling becomes min: the curve stays within $\log 2/|\log s|$ of its larger term (shaded)



(b) the cotangent cone at a boundary law: a tied act is injection-proof iff its utility gap lies in C_x^*